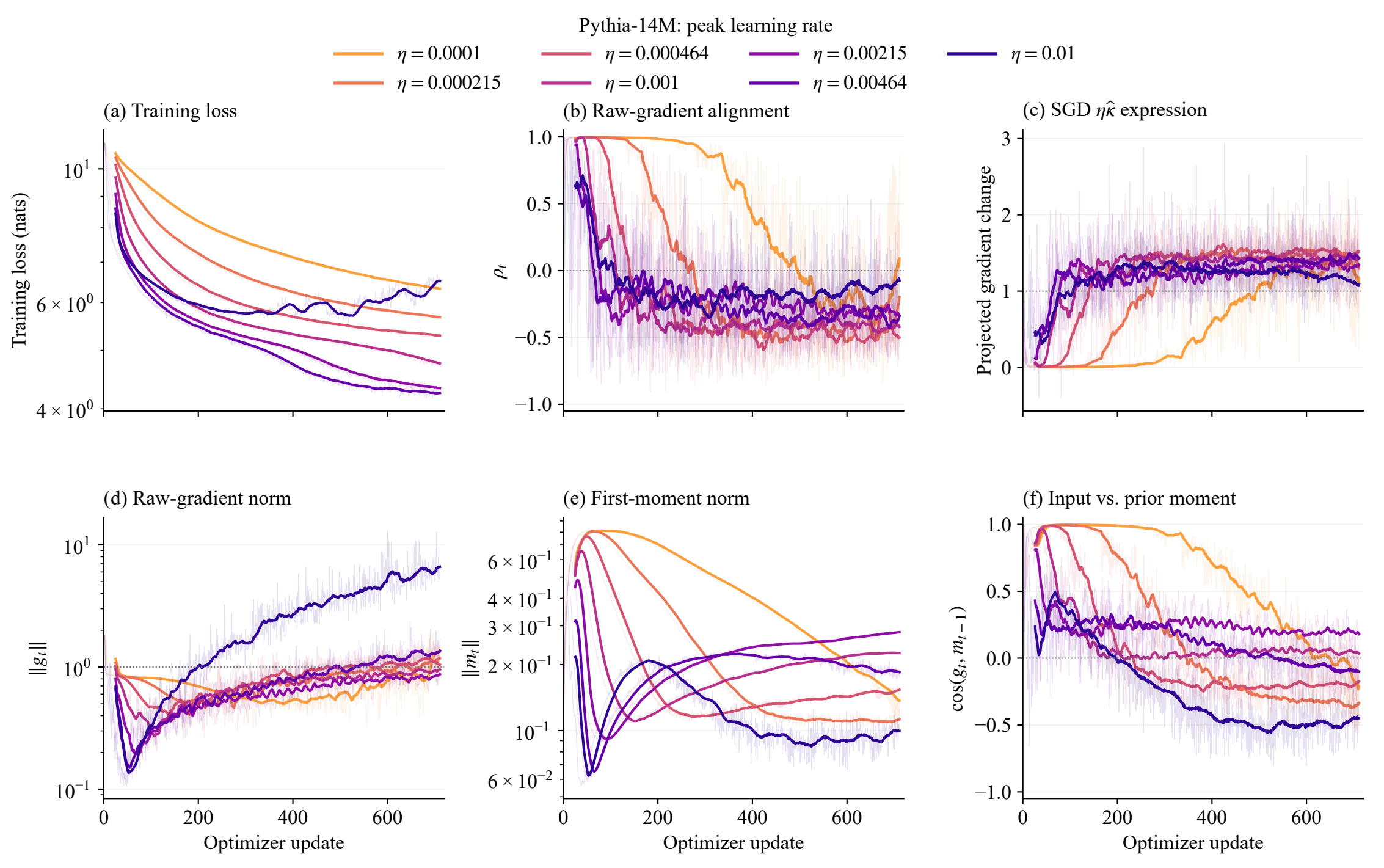

Pythia-14M: peak learning rate
$\eta = 0.0001$
$\eta = 0.000215$
$\eta = 0.000464$
$\eta = 0.001$
$\eta = 0.00215$
$\eta = 0.00464$
$\eta = 0.01$
(a) Training loss
Training loss (nats)
10^1
6×10^0
4×10^0
(b) Raw-gradient alignment
ρ_t
1.0
0.5
0.0
−0.5
−1.0
(c) SGD $\eta\hat{\kappa}$ expression
Projected gradient change
3
2
1
0
(d) Raw-gradient norm
$\|g_t\|$
10^1
10^0
10^{-1}
0
200
400
600
Optimizer update
(e) First-moment norm
$\|m_t\|$
6×10^{-1}
4×10^{-1}
3×10^{-1}
2×10^{-1}
10^{-1}
6×10^{-2}
0
200
400
600
Optimizer update
(f) Input vs. prior moment
$\cos(g_t, m_{t-1})$
1.0
0.5
0.0
−0.5
−1.0
0
200
400
600
Optimizer update